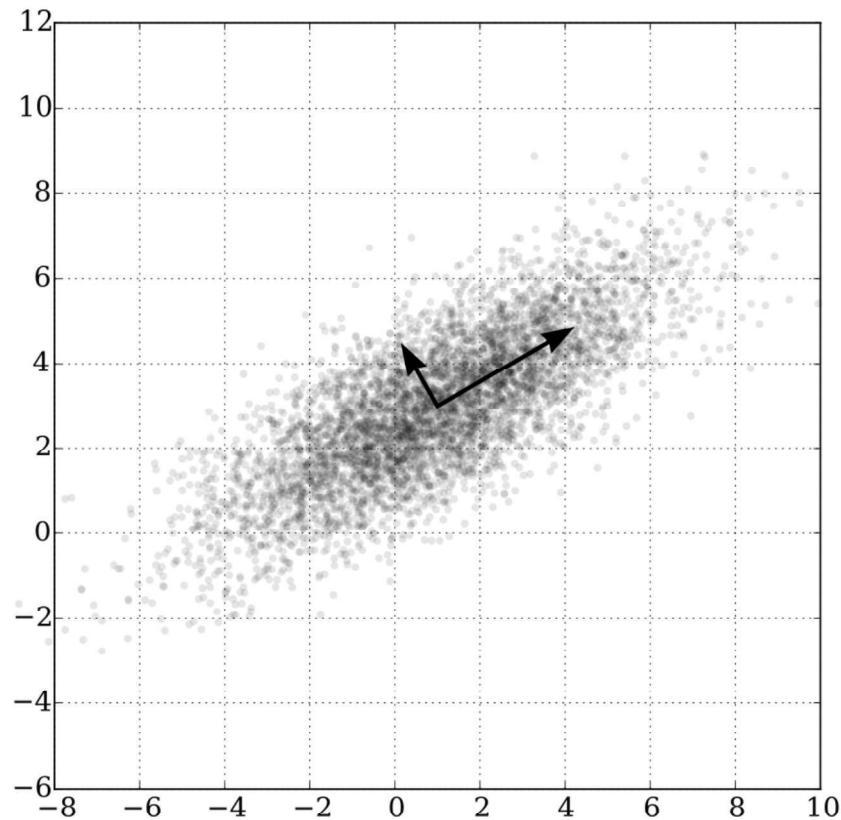


PCA

- creates new variables using linear combinations of old variables
- is designed to create variables that are independent of one another
- also manages to tell us how important each of these new variables are
- this “importance”, helps us to choose how many variables we will use



- Scale the data and compute the covariance matrix
- Break the covariance matrix into magnitude and direction. Eigen Vectors and the Eigen Values of the covariance matrix can be thought of as the natural axis/directions and magnitudes along those axis, of the data
- The eigen values also can be used to calculate the percentage of variation explained by each component
- Sort in the eigen values in descending order and calculate the cumulative percentage of variation explained
- Pick the number of principal components you will use
- Transform to new variables

